

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	BE CSE AI / 2024	Date:	13-08-2026

Code of Conduct

/ NARAPPAGARI SRVAYA (192472192)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SRAVYA

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

- Analyze the semantic representations and identify the action-object relationships.
- Determine which query contains semantic interpretation errors.
- Evaluate how meaning representation affects chatbot decision accuracy.
- Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules. Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
 2. Analyze which products are currently available.
 3. Infer whether Gear production is affected by maintenance activities.
 4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.
3. Word Sense Disambiguation in E-Commerce Search Engines. An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Semantic Representation in Customer Support Chatbots

1. Action–Object Relationships

Query	Semantic Representation	Action	Object
Q1	ACTIVATE(Roaming, Customer)	Activate	Roaming
Q2	DEACTIVATE(CallerTune, Customer)	Deactivate	Caller Tune
Q3	QUERY(DataBalance, Customer)	Query/Check	Data Balance
Q4	ACTIVATE(5GService, Customer)	Activate	5G Service

The semantic representation correctly identifies **what action the customer wants to perform and on which service/object**.

2. Semantic Interpretation Error

The actual and predicted intents are:

- Q1 → Activate Roaming → **Correct**
- Q2 → Deactivate Caller Tune → **Correct**
- Q3 → Check Data Balance → **Correct**
- Q4 → Enable 5G Service → **Correct**

However, the table also shows “**Activate Caller Tune**” as a predicted intent, which does not match the query “**Deactivate caller tune service.**”

Therefore, the error is the interpretation of **Q2**, if “Activate Caller Tune” is the prediction associated with Q2.

3. Effect on Chatbot Accuracy

Correct semantic representation helps the chatbot:

- Understand the user's intention.
- Identify the correct service.
- Perform the correct action.
- Give an appropriate response.

Incorrect representation can cause the chatbot to perform the **opposite or wrong operation**, reducing accuracy and customer satisfaction.

4. Improvements

- Use context-aware NLP models.
- Use telecom-specific training data.
- Include synonyms such as *enable*, *activate*, *turn on*.
- Handle negation carefully, such as *deactivate*, *disable*, *stop*.
- Use intent classification with entity extraction.
- Use RAG/knowledge bases for telecom service information.
- Continuously retrain the system using real customer interactions.

2. First-Order Predicate Calculus for Smart Manufacturing

1. Production Data in First-Order Predicate Calculus

Given:

- $M1 \rightarrow \text{Active}$
- $M2 \rightarrow \text{Active}$
- $M3 \rightarrow \text{Maintenance}$
- $M4 \rightarrow \text{Active}$

Represented as:

- **Active(M1)**
- **Active(M2)**
- **Maintenance(M3)**
- **Active(M4)**

Rules:

1. $\forall x \text{ Active}(x) \rightarrow \text{Producing}(x)$
2. $\forall x \forall y [\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)]$
3. $\forall x \text{ Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

2. Products Currently Available

From the rules:

- M1 is Active $\rightarrow$ M1 is Producing.
- M2 is Active $\rightarrow$ M2 is Producing.
- M4 is Active $\rightarrow$ M4 is Producing.
- M3 is under Maintenance $\rightarrow$ M3 is **not Producing**.

A specific product can be declared **Available** only when there is a fact of the form:

Produces(x,y)

together with **Active(x)**.

Since no Produces(x,y) facts are provided, no specific product can be conclusively identified as available.

3. Effect of Maintenance on Gear Production

If **M3 produces Gear**, then:

Produces(M3, Gear) $\wedge$ Maintenance(M3)

Maintenance implies:

Maintenance(M3) $\rightarrow \neg$ Producing(M3)

Therefore, **Gear production by M3 is affected/stopped.**

4. Effectiveness of Predicate Logic

Predicate logic is useful because it:

- Represents machines and their states clearly.
- Allows automatic rule-based reasoning.
- Detects production problems.
- Supports decision-making.
- Represents relationships between machines and products.

However, it may be less effective when manufacturing data is **uncertain, incomplete, or continuously changing**. In such cases, it can be combined with machine learning and probabilistic methods.

3. Word Sense Disambiguation in E-Commerce Search Engines

1. Correct Word Sense

Query	Correct Sense	Evidence
Apple accessories	Technology Brand	Clicked result: iPhone Charger
Mouse wireless	Computer Device	Clicked result: Bluetooth Mouse
Java tutorial	Programming Language	Result: Coding Lessons
Python course	Programming Language	Result: Software Development Training

2. Semantic Cues

Important cues include:

- **Accessories + iPhone Charger** → Apple technology brand.
- **Wireless + Bluetooth Mouse** → computer mouse.
- **Tutorial + Coding Lessons** → Java programming language.
- **Course + Software Development Training** → Python programming language.

The search engine uses the **surrounding words, clicked results, and search context** to determine the intended meaning.

3. Effect of Incorrect Sense Selection

Incorrect WSD can lead to:

- Irrelevant search results.
- Poor product recommendations.
- Lower click-through rate.
- Reduced customer satisfaction.
- Lower sales/conversion rate.

For example, interpreting “**Apple accessories**” as fruit-related products would produce completely irrelevant results.

4. Industrial-Scale WSD Strategy

An e-commerce system can use:

1. Context-aware transformer models such as BERT.

2. Query history and user behavior.
3. Product-category information.
4. Click and purchase history.
5. Entity recognition.
6. Knowledge graphs.
7. Continuous feedback from user interactions.

This combination can improve search and recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare

1. How Syntax Helps Semantic Roles

Syntax identifies the relationship between words in a sentence.

Examples:

- **Doctor prescribed medicine to patient.**
 - Doctor → Agent
 - Medicine → Theme/Instrument
 - Patient → Recipient
- **Patient reported severe headache.**
 - Patient → Agent/Experiencer
 - Headache → Symptom
- **Nurse monitored patient continuously.**
 - Nurse → Agent
 - Patient → Patient/Theme
- **Medicine reduced blood pressure.**
 - Medicine → Cause/Agent
 - Blood pressure → Theme

The **Subject–Verb–Object structure** helps the NLP system determine who performed an action, what was affected, and who received it.

2. Are the Assigned Roles Appropriate?

Most roles are appropriate, but “**Medicine** → **Instrument**” is not the best semantic role for:

Doctor prescribed medicine to patient.

Here, **Medicine** is better represented as the **Theme**, while **Patient** is the **Recipient**.

So:

Doctor = Agent → **Medicine = Theme** → **Patient = Recipient**

3. Errors from Incorrect Parsing

Incorrect syntactic parsing can cause:

- Wrong identification of doctor and patient.
- Incorrect medication information.
- Wrong diagnosis interpretation.
- Incorrect identification of symptoms.
- Errors in medical records.
- Incorrect clinical decision support.

For example, confusing “**Doctor prescribed medicine to patient**” with the reverse relationship could lead to serious information errors.

4. Improvements

Healthcare NLP systems should use:

- Medical-domain language models.
- Dependency parsing.
- Semantic role labeling.
- Medical dictionaries/ontologies.
- Context-aware NLP models.
- Human validation for critical information.
- Training datasets containing real medical sentence structures.

Conclusion: Syntax provides the structural foundation, while semantic analysis converts that structure into meaningful roles such as **Agent, Patient, Recipient, Theme, and Symptom**.